

WASP-35b

R-1576

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_171230 · created 2026-08-01T23:16:06+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458117.6405 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1304 +/- 0.0077
Transit depth (Rp/Rs)^2	1.7 +/- 0.2 [%]
Orbital Inclination (inc)	89.81 +/- 0.89
Airmass coefficient 1 (a1)	1.145 +/- 0.01
Airmass coefficient 2 (a2)	-0.1032 +/- 0.0021
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	None
Optimal Aperture	4.54
Optimal Annulus	13.4
Transit Duration (day)	0.1338 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1304 ± 0.0077 vs 0.1241 (deviation 0.7σ)
Duration fit vs published	0.1338 d vs 0.1317 d (deviation 1.22σ)
Mid-time O–C	-3.4 min
Depth SNR	14.5
Residual scatter	0.87 %

Light curve

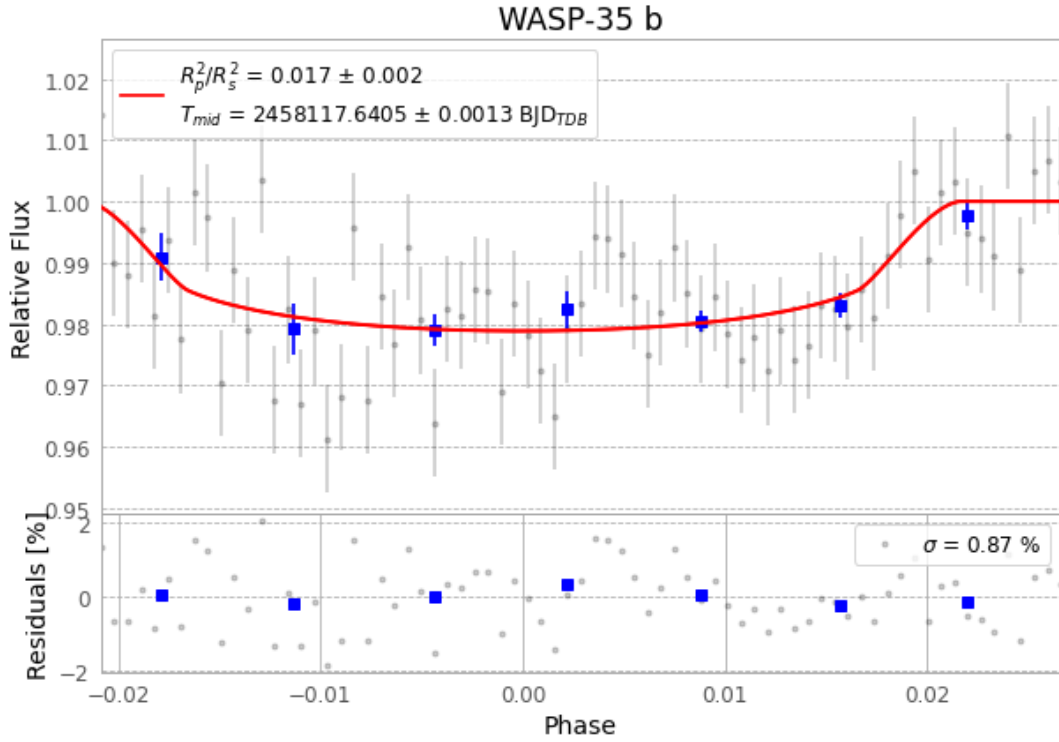


Figure 1 — FinalLightCurve_WASP-35 b_2017-12-29.png (SHA-256 8d8cd8c27b18fa9c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2453cb5a07c9561b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-35171230013912.FITS → WASP-35171230051512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-171230.log (aadf08a978d9...), FinalLightCurve_WASP-35 b_2017-12-29.png (8d8cd8c27b18...), FinalLightCurve_WASP-35 b_2017-12-29.csv (06e27796a8ba...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:16Z.